

Leveraging an October Surprise to Estimate Coattail Effects

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March 26, 2025

Abstract

Does a presidential candidate's popularity impact her party's down-ballot success? So-called "coattail effects" can be challenging to identify: it is difficult to distinguish between voters' opinion of a prominent candidate and her political party. To overcome this challenge, I exploit a shock to candidate popularity created by late-election information in the 2016 presidential election. Results indicate that exposure to FBI Director James Comey's letter to Congress significantly increased Republican presidential vote share. Using an instrumented difference-in-differences model, I find that a one percentage point increase in Republican presidential vote share increases down-ballot Republican vote share by 0.86 percentage points.

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I am grateful for comments from Brian Beach, Peter Bils, Patrick Flynn, Mattias Polborn, Tucker Smith, Melissa Spencer, and Lesley Turner. I also thank seminar and conference participants from Vanderbilt University, Virginia Commonwealth University, the Liberal Arts Labor and Public Conference, and the June 2024 CSWEP CeMENT workshop.

1 Introduction

Political parties act as a coordination mechanism through which voters are mobilized, resources raised, and policies passed in the name of advancing common ideological goals. They also lower informational costs to voters: rather than taking the time to evaluate every candidate each election, a voter can interpret a party affiliation as a signal of a candidate’s values and objectives. This generates the possibility for electoral spillovers between candidates, especially from ones with high visibility to those with less. These spillovers are known as “coattail effects”: a candidate has coattail influence if down-ballot candidates in her same party receive votes they otherwise would not have received, save for her popularity [Miller, 1955]. Coattail effects, though widely believed to exist, can be challenging to quantify: the popularity of a single candidate is difficult to parse from the public’s sentiment toward her political party. In this paper, I provide the first causally identified estimates of presidential coattail effects.

I estimate presidential coattail effects using a natural experiment created by late-election information in the 2016 U.S. presidential election. On October 28, 2016, then-FBI Director James Comey sent a letter to Congress regarding Democratic presidential candidate Hillary Clinton. Comey announced the discovery of additional emails which pertained to an investigation into Clinton’s use of a private email server while serving as Secretary of State – an investigation he previously testified was complete. The letter reflected negatively on Clinton, but variation in early voting laws across states meant voters had differing abilities to react to the information. Many people in states with early voting had already cast their ballots. In my central identification strategy, I compare election outcomes in counties with no-excuse early voting available before 10/28 to those in counties where no-excuse early voting was either unavailable or began after 10/28, relative to election outcomes in 2012.

First, I show that the Comey letter impacted the 2016 presidential election: exposure

to the letter decreased the Democratic presidential vote share by 2 percentage points and increased the Republican presidential vote share by 3.2 percentage points. This increase is larger than the vote margins of eight states in 2016: of these eight, four did not have early voting ahead of October 28th, and three of those four (Florida, Michigan, and Pennsylvania) broke in favor of Republican candidate Donald Trump.¹ Using this as a first stage, I then use an instrumented difference-in-differences model to estimate coattail effects. I find that a one percentage point increase in the Republican presidential vote share led to a 0.86 percentage point increase in the down-ballot Republican vote share.

We might be concerned that these findings reflect differential political trends of early voting and non-early voting states rather than the impact of the Comey letter. To address these concerns, I show that other items on states' ballots are unrelated to the timing and availability of early voting with a balance test.² I find no relationship between treatment and the presence of toss-up Senate or gubernatorial races, the number and type of ballot initiatives, the presence of a third-party presidential candidate on the ticket, or straight-ticket voting, supporting the parallel trends assumption.

To provide additional support for my results, I introduce two complementary empirical designs which limit to within-state or early voting state-only comparisons. In the first, I make use of data on partisan turnout during early voting periods in Nevada and North Carolina. Each state reports data on turnout by date throughout early voting periods. This provides for a natural comparison between pre-October 28th and post-October 28th early voters. I find that the letter decreased registered Democrats' share of early turnout by 2.3 and 0.9 percentage points in North Carolina and Nevada, respectively.

In the second complementary design, I limit my analysis to six early voting states which

¹Of the four that did have early voting ahead of the letter's release, three (Minnesota, Nevada, and Maine) broke in favor of Clinton.

²In particular, I run variations of my difference-in-differences design using other characteristics of states' ballots as outcomes of interest.

report votes cast on Election Day vs. early voting days separately.³ This allows me to compare partisan vote shares of early voters and Election Day voters to assess the Comey effect, ruling out potential confounding differences between states with and without no-excuse early voting. These states had varying amounts of early voting available ahead of the letter’s release. I expect to observe larger effects in states that had more (vs. less) days of early voting available prior to 10/28, as those states presumably had a larger proportion of voters casting their ballot with no knowledge of the letter. I use a difference-in-differences design with continuous treatment, where the number of days of early voting available before 10/28 is the treatment dose. In particular, I find that in a county with 10.17 days of early voting ahead of the Comey letter (the average number of days of early voting prior to 10/28 across all states with no-excuse early voting in 2016), exposure to the letter decreased Democratic presidential vote share by 2.03 percentage points and increased Republican presidential vote share by 1.02 percentage points.

While theoretical approaches have predicted electoral spillovers between different levels of government in a federalist system, this paper provides the first causally identified empirical estimates of presidential coattail effects. Formal theoretical treatments include [Zudenkova \[2011\]](#) and [Halberstam and Montagnes \[2015\]](#). [Zudenkova \[2011\]](#) documents that coattail effects arise if we assume incumbents prefer their co-partisans to win races in different levels of government.⁴ [Halberstam and Montagnes \[2015\]](#) document that senators first elected in presidential years are more ideologically extreme than those first elected in midterms. To explain this observed pattern, they devise a model which assumes the existence of presidential coattail effects and shows how such effects can lead to ideological differences in senators

³The six states are Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina.

⁴Knowing this, retrospective voters (meaning they vote based on an incumbent’s performance) jointly evaluate co-partisans, providing for two-sided coattails (that is, spillovers both up and down the ballot).

depending on when they run.⁵

However, it is difficult to validate these theories: there are inherent complications in separating a particular candidate’s popularity from that of her political party.⁶ How might we distinguish coattail effects from party politics? Coattail effects are defined as the impact of a candidate’s personal popularity on her co-partisans’ success elsewhere on the ballot. Thus, proper identification of such an effect requires some exogenous shock to a candidate’s popularity – importantly, separate from the favorability of her party at large – and variation in voters’ experience of that shock.

Meredith [2013] exploits the excess support granted by (geographically) nearby voters to estimate gubernatorial coattail effects, finding that a one percentage point increase in gubernatorial vote share leads to 0.1-0.2 percentage point increases in vote shares for partisan allies running for secretary of state and attorney general. Many recent papers use close-election regression discontinuity designs to estimate the effect of a co-partisan winning a prior election in a different level of government. These have yielded mixed results on the existence and direction of such “incumbency spillovers,” depending on setting, the levels of government, and the direction [Hainmueller and Kern, 2008, Broockman, 2009, Folke and Snyder, 2012, Ade and Freier, 2013, Feierherd, 2020, Ventura, 2021, Savu, 2024].⁷

It is possible that each of these designs could be capturing not just an increase in the personal popularity of a candidate, but also an amplified opportunity for party rhetoric.

⁵It should be noted that this pattern of differing senator ideologies may have other explanations. For example, it could be that parties run more extreme Senate candidates to generate additional excitement for the party in presidential years. The documented correlation is between senator ideology and the *presence* of a presidential election, not co-partisan presidential success, and thus could be a result of strategic choices by both parties in presidential years, independent of their presidential candidates’ popularity.

⁶Most prior work consists of observational studies which attempt to control for determinants of down-ballot vote shares [Campbell and Sumners, 1990, Mondak, 1993, Flemming, 1995, Mattei and Glasgow, 2005] or structural models [Kramer, 1971, Ferejohn and Calvert, 1984] which have proven to be sensitive to model specification [Fair, 2009]. See Meredith [2013] for a detailed discussion of the difficulties of identifying coattail effects.

⁷Several of these papers estimate “reverse coattails,” which refer to spillovers from down-ballot candidates to their fellow party members higher up on the ticket; for example, from a Democratic mayoral candidate to a Democratic U.S. House candidate.

Geographically proximate candidates and incumbents alike have increased opportunities to positively advertise their own political party, complicating the distinction between coattail effects and party politics. Instead, the Comey letter (or more broadly, late-election information) provides a plausibly exogenous shock to candidate popularity alone.

Late-election information offers the popularity shock needed to estimate presidential coattails, and disparate state-level early voting laws supply the necessary variation. Key to my empirical design is the notion that convenience voters and Election Day voters face fundamentally different information environments [Meredith and Malhotra, 2011]. My identification strategy is similar to ones used in Montalvo [2011], which shows that a terrorist bombing in Spain negatively impacted the performance of the incumbent party in the 2004 general election which took place three days later, and Graham and Svolik [2020], which shows that a Montana U.S. House candidate’s assault of a journalist the night before an election hurt his performance with moderate voters on Election Day. This paper is the first to use late-election information to 1) analyze its impact in a U.S. presidential race and 2) identify coattail effects.

The paper is also related to work on the impact of information and media in elections [Couttenier et al., 2024, Ash and Galletta, 2023, Wang, 2021, Morton et al., 2015]. Some work has been done on the impact of the Comey letter, specifically: there is some evidence that the letter had impacts in both polling and electoral prediction markets [Halcoussis et al., 2020, Silver, 2017]. McKee et al. [2019] points out that in each of the seven swing states that Trump won, he lost the sum of the votes cast early in-person or by mail; this paper builds on this by using variation in early voting to provide the first quasi-experimental analysis of the Comey letter.

Identifying the presence and magnitude of coattail effects furthers our understanding of voter behavior and has significant strategic implications for politicians. While coattail effects

are likely not a determinant of down-ballot candidates’ party affiliation,⁸ they may influence how closely such candidates choose to align themselves with up-ballot ones. Presidential coattail effects may even influence *when* down-ballot candidates run: they may wish to obtain an electoral boost from a popular co-partisan or avoid being dragged down by an unpopular one. Coattails have important ramifications for presidential candidates, too: strong spillover effects into state-level races should increase a president’s legislative efficacy once in office. Finally, this paper’s finding that late-election information can impact presidential and down-ballot races suggests that the incentives for the strategic release of late-election information may be even larger than previously thought [Gratton et al., 2018].

2 Setting

In this section, I provide background information on convenience voting in the United States – key to my identification strategy – and on James Comey’s letter to Congress.

2.1 Convenience Voting in the United States

Thirty-five states began convenience voting before October 28th, 2016, the day of the Comey letter’s publication. A total of thirty-seven states practiced no-excuse convenience voting at the time: see Figure 1 for statewide variation (note that Florida and Oklahoma had no-excuse convenience voting, but their voting periods did not open until *after* October 28th). Of those thirty-seven, three states used universal vote-by-mail, meaning ballots were mailed to all registered voters.⁹ The rest of the thirty-seven had early in-person voting options, where citizens could cast their ballot in-person up to more than six weeks ahead of Election

⁸Though there is a recent uptick in party-switching amongst state lawmakers in the U.S. [Crampton, 2023].

⁹These three states were Colorado, Oregon, and Washington; nearly 80% of counties in Utah were also using universal vote-by-mail in 2016 [Thompson et al., 2020].

Day.¹⁰

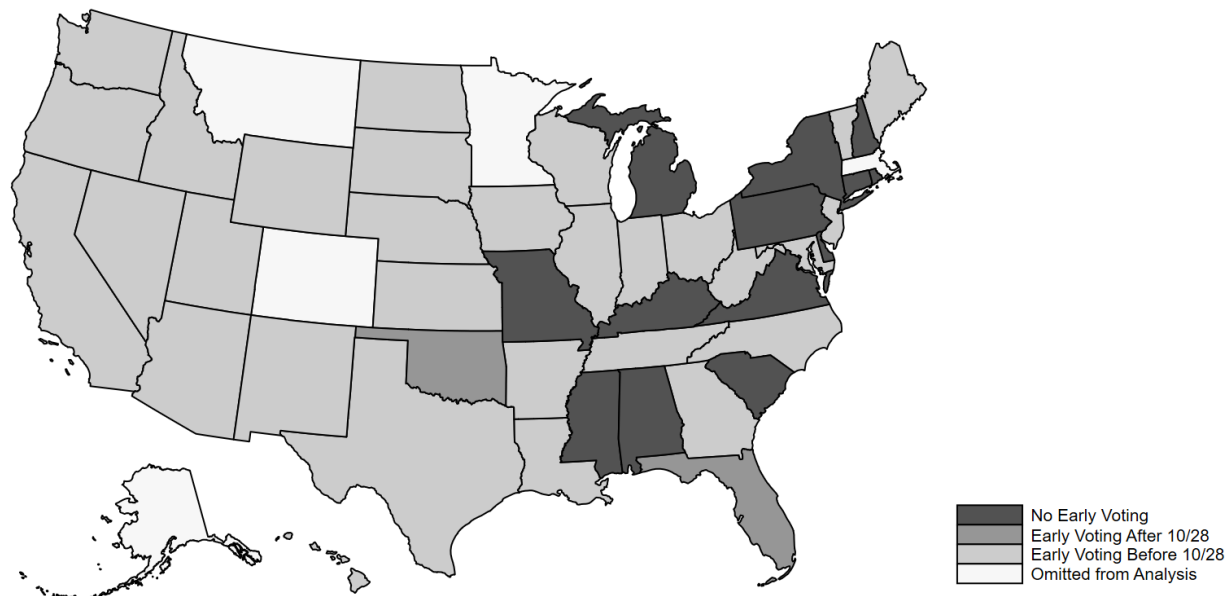


Figure 1: This map displays variation in convenience voting laws for U.S. states in 2016. Notes: Four states (Colorado, Massachusetts, Minnesota, and Montana) made changes to their convenience voting laws between 2012 and 2016; I omit them from my analysis. I also omit Alaska, as it does not report election results by county.

See Table 1 for summary statistics on counties in states with convenience voting and states without convenience voting. A potential concern with comparing the two groups is that there may be underlying qualities which differ between the two and are somehow correlated with treatment. That is, if one group was more or less responsive to the Comey letter due to differential baseline characteristics. However, as Table 1 shows, the two groups possess comparable baseline demographic, economic, and political attributes. The two groups appear quite similar along most dimensions; the largest difference is in the unemployment rate (there, the groups differ by 0.424 standard deviations).

¹⁰The earliest early voting start date in 2016 was September 21st in Wisconsin.

Table 1: This table displays summary statistics in 2012 for counties in the United States, broken into states with convenience voting and counties without convenience voting.

	Convenience Voting	No Convenience Voting	Difference (Std. Devs.)
18+ Population	78,171.43	77,744.84	0.002
Unemployment Rate	7.554	8.74	0.424
Population Density	136.341	410.587	0.184
Median Age	40.258	40.279	0.004
# Men Per 100 Women	100.823	98.334	0.221
% Non-white	15.434	18.461	0.183
Median Income	45,540.93	44,108.41	0.122
% with Bach. or Higher	19.024	19.148	0.015
Dem/Rep. House Margin*	-0.234	-0.158	0.241
Dem. Vote Share in Pres. Race	0.368	0.417	0.331
Turnout	0.549	0.568	0.2
Count	2090	801	

Notes: The means of the convenience voting and no convenience voting groups are reported, along with the difference in terms of the standard deviation of the entire sample, for reference. For example, the difference in 18+ population between the groups is 0.002 of one standard deviation (measured from the entire sample). The variable denoted with an asterisk is averaged over three U.S. House election cycles: 2008, 2010, and 2014. All other variables are taken from 2012. Colorado, Massachusetts, Minnesota, and Montana are not included, as they changed convenience voting laws between 2012 and 2016; Alaska is not included, as it does not report election results by county. Population density is calculated by dividing a county’s 18+ population by its land area in square miles (as reported by the 2010 Census). The % with Bach. or Higher variable refers to the percent of those aged 25+ with a bachelor’s or more advanced degree.

2.2 The Comey Letter as Late-Election Information

Late-election information, broadly speaking, is revealed to voters in the weeks or days preceding Election Day and has the potential to change voters’ minds. This information can take the form of something already structured within the political system, such as presidential debates or campaign stops, or a spontaneous revelation, known as the “October surprise” in American politics. Interested parties certainly have incentives to release information with the intent to impact election outcomes as Election Day looms.¹¹ However, this late-election information, whether spontaneous or strategically released, can nevertheless “surprise” the

¹¹Gratton et al. [2018] present this as a tradeoff between credibility and scrutiny: good signals of candidate quality are released earlier, as this conveys credibility of the information, but poor signals will be released later, since they would not hold up to the public’s scrutiny if examined for too long.

public and might induce changes in voter behavior.

There are many recent examples of late-election information in the United States: in 2020, news of an extramarital affair and a subsequent investigation by the U.S. Army Reserve of Cal Cunningham, a Democrat running for a U.S. Senate seat in North Carolina, occurred less than one month away from Election Day [Robertson, 2020]. In the 2003 California gubernatorial recall election, Republican Arnold Schwarzenegger replaced incumbent Democratic Governor Gray Davis, winning by a margin of 968,491 votes [Ballotpedia, c]. The *Los Angeles Times* reported on the Friday before the election that six women had accused Schwarzenegger of sexual misconduct [Cohn et al., 2003] after “more than three million of ten million voters had already cast their ballots” [Gronke et al., 2008]. In 2000, news of George W. Bush’s 1976 DUI arrest was reported on by media outlets four days before Election Day [Balz, 2000].

In this paper, I focus on a particular example of late-election information: then-FBI Director James Comey’s October 28, 2016 letter to Congress. Comey announced in the letter that the FBI had discovered more emails which seemed pertinent to the investigation of Hillary Clinton’s use of a private email server during her time as Secretary of State (an investigation Comey had previously testified was complete). The letter was sent to Congress just eleven days before the presidential election between Republican candidate Donald Trump and Democratic candidate Hillary Clinton [Perez and Brown, 2016].

The last month or so of the campaign leading up to Election Day on November 8th was quite dynamic in terms of late-election information. On October 7th, *The Washington Post* obtained a video in which “Donald Trump bragged in vulgar terms about kissing, groping and trying to have sex with women during a 2005 conversation caught on a hot microphone” [Fahrenthold, 2016]. This prompted several prominent Republicans to condemn Trump; some even suggested he withdraw his candidacy [Wellford, 2016]. On that same day, the Office of the Director of National Intelligence and the Department of Homeland Security released a

statement accusing Russia of interfering with the election, and media organization WikiLeaks began releasing emails belonging to Clinton campaign chairman John Podesta, many of them “embarrassing for Clinton” [Cohen, 2017]. These releases continued on a near-daily basis throughout the rest of the campaign, but none seemed to generate a nation-wide shock in the way that 1) the aforementioned tape, 2) the beginning of the WikiLeaks releases, or 3) the Comey letter did.

I choose to focus on the Comey letter as the late-election information of interest in the election. The events of October 7th negatively impacted both candidates, prompting muddled behavioral implications for voters. In comparison, the Comey letter (and thus October 28th) clearly represented negative information for Clinton alone. Additionally, not as many people had yet had the opportunity to vote by October 7th, making for a smaller control group: just ten states began convenience voting before October 7th.¹²

As mentioned above, thirty-five states began convenience voting before October 28th. Although concerns over Clinton’s use of the email server had already been raised and an investigation completed, this letter and its implication of a re-opened investigation may have legitimized some voter’s concerns about the candidate. It certainly represented an information shock of interest to many voters; see Figure 2 in the appendix for Google Trends data, which shows a large spike in searches for “Comey” on 10/28/16. The election was still eleven days away, yet many had already cast their ballot before they learned of this investigation. Their hands were now tied; would they have voted differently had they waited until Election Day, or perhaps abstained altogether?

¹²In a robustness check, I drop these ten states from my control group to ensure that all counties, both treated and control, had knowledge of the events of October 7th prior to voting: see Section A.4.

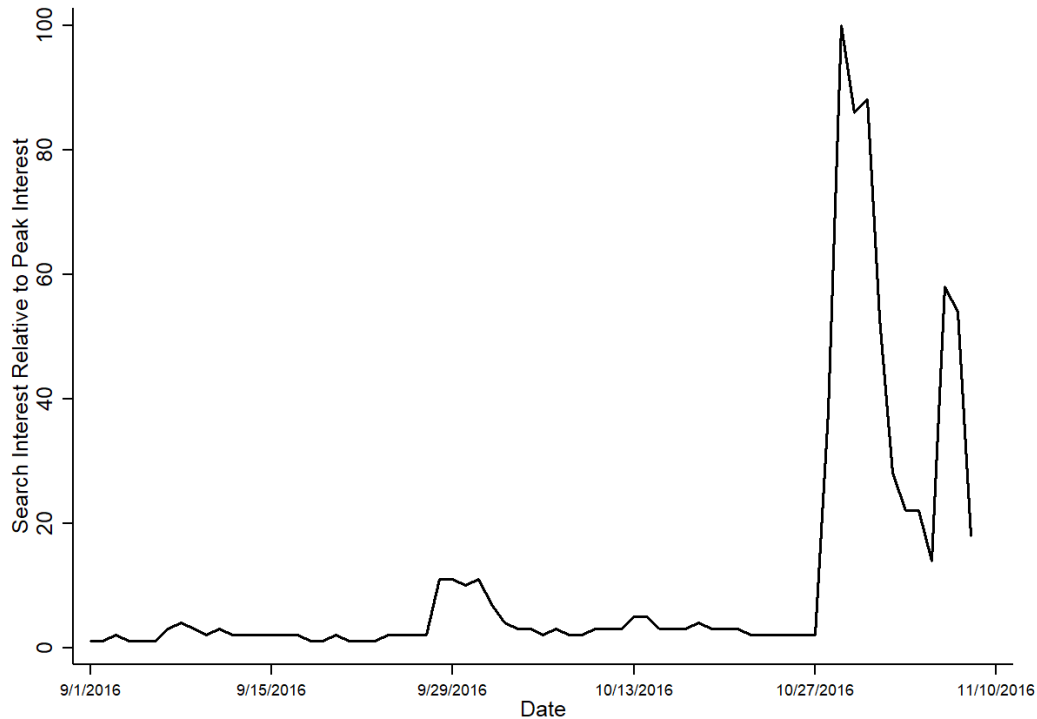


Figure 2: This figure shows Google Trends data for searches for “Comey” in the United States between 9/1/2016-11/8/2016. A value of 100 is peak popularity for the term.

3 Data & Methodology

In this section, I present the data and methodology used in my analysis. I make use of three complementary empirical strategies in this paper. The main design, which relies on cross-state variation in the availability of early voting, allows me to examine how the Comey letter impacted voter turnout and partisan presidential vote shares. In addition, I study the performance of state-level candidates to determine if the letter had spillover effects based on party affiliation. The second identification strategy leverages a unique feature of two states’ (Nevada and North Carolina) elections data – disaggregated partisan early voting turnout data – to compare pre-Comey letter early voters to post-Comey letter early voters. The third specification compares votes cast on Election Day vs. during the early voting period

in the six states which report these totals separately (Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina). The latter two designs allow me to rule out confounding differences between early voting and non-early voting states, but at the cost of a smaller sample size.

3.1 Data

I obtain data on convenience voting laws and timing from the Elections & Voting Information Center for the 2012 election [Hicks, 2012] and Ballotpedia for the 2016 election [Ballotpedia, b] (Accessed: Sep. 2021) and use it to define treatment in the central difference-in-differences design: states with early voting windows which open before 10/28/16 are control states and those without early voting windows open before 10/28/16 are treatment states (this includes states which did not have early voting at all). Treatment is therefore defined by the *lack* of ability to vote early without an excuse before the letter’s release, as voters in these states would have been less constrained in their means to respond to the letter in the ballot box.

I collect county-level presidential, congressional, and gubernatorial election returns (vote totals for each candidate) for 2012 and 2016 from Dave Leip’s Atlas of Elections [Leip] (Accessed: Apr. 2022 and Nov. 2023) and counts of 18+ population for each county in 2012 and 2016 from the Survey of Epidemiology and End Results (SEER). I use these counts as voting-age population for calculating turnout rates. For the two complementary designs, I collect county-level vote returns and turnout reports for 2012 and 2016 from each state’s election administrator. Finally, I obtain county-level control variables from the U.S. Census Bureau’s American Community Survey 5-year estimates, the U.S. Bureau of Labor Statistics, and the U.S. Census Bureau’s Census of Population and Housing.

3.2 State-Level Variation in Convenience Voting

In the main empirical specification, I categorize two types of counties as treated: counties in states without no-excuse early voting and counties in states where early voting did not open until after October 28th, 2016. The treatment turns on with the Comey letter (only in 2016 and not in 2012). I use counties in states with early voting which opened before October 28th, 2016 as control units, as voters in these states would have been least able to respond to the information. Here, the variation of interest comes from the availability of no-excuse early voting, something set independently by each state.¹³

The identifying assumption here is that, absent the Comey letter, treatment and control counties would have trended similarly in the outcomes of interest (turnout rate and partisan vote shares). This design also requires that any late-election information in 2012 did not affect political behavior. I estimate

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ct}, \quad (1)$$

where T_c takes a value of 1 for counties in states without early voting and states where early voting did not open until after October 28th, 2016 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 otherwise.

Outcomes of interest y_{ct} for county c in year t include turnout rate, vote shares for Democratic and Republican presidential candidates, and average vote shares for Democratic and Republican state-level candidates. I calculate turnout rate as a proportion of the voting-age population (i.e., the 18+ population); presidential vote shares are proportions of total ballots cast for the president. To determine the average partisan vote shares for state-level candidates, I average together the Democratic (or Republican) vote shares in U.S. House,

¹³I exclude from my analysis any state which changed its convenience voting laws between 2012 and 2016 (Colorado, Massachusetts, Minnesota, and Michigan).

Senate, and gubernatorial elections.¹⁴ I weight by voting-age population, since 1) I expect more accurate measurement of rates in counties with larger populations and 2) treatment effects could be correlated with population (insofar as how population might be related to political ideology, etc.).

3.2.1 Support for Identification

One potential threat to identification with the comparison of early voting to non-early voting states is the possibility that these two sets of states may have been experiencing different political trends. If this were true, we might expect to see it directly reflected in those states' ballots. Perhaps one set of states had a separate reason to swing left or right in 2016: toss-up down-ballot races, ballot initiatives, etc. I collect data on states' 2012 and 2016 ballot characteristics in four broad categories: toss-up Senate or gubernatorial races, the number and type of ballot initiatives, third-party presidential candidates on the ballot, and straight-ticket voting.^{15,16}

To provide support for the parallel trends assumption, I perform a balance test using the variables described above. Results are shown in Section A.1; I find that there is no relationship between “treatment” (exposure to the Comey letter, proxied by a lack of early voting available before 10/28) and any of these items on the ballot. I further address the “differing trends” concern with two complementary empirical specifications, introduced below.

Note that counties in Florida and Oklahoma may be a superior “treated” group, as both Florida and Oklahoma had no-excuse early voting periods that did not open until after 10/28/16. In a specification using only Florida and Oklahoma counties as treated units,

¹⁴I exclude unopposed elections, as vote shares would not reflect any spillover effects in these cases.

¹⁵Toss-up races are the most competitive races of the election cycle, where neither party is predicted to have a slight edge.

¹⁶Straight-ticket voting allows a voter to vote for all of their chosen party's candidates for each partisan office on the ballot with a single mark of their ballot.

both treated and control voters live in states with early voting: the difference is just in when early voting is offered. Results are robust to such a refinement of the design (see Section A.2).

Identification also requires that we assume that any late-election information in 2012 did not affect turnout or partisan vote shares. Some may view Hurricane Sandy, which made landfall in the U.S. on October 29th, 2012, as late-election information which benefited President Obama (since his handling of the storm was positively regarded by both parties). See Section A.3 for a robustness check; results are robust to leaving out counties in Connecticut, New Jersey, and New York, which had the largest populations in storm surge zones during Hurricane Sandy [Center for International Earth Science Information Network, 2012].

It is possible that my main empirical approach picks up larger dynamics within the election, rather than voters' response to the singular shock of the Comey letter. Note that some control counties had early voting available prior to October 7th, another important day in the election cycle. In Section A.4, I show results are robust to the exclusion of all counties which had early voting available prior to 10/7. Even still, perhaps Clinton and her fellow Democrats would have faced a downturn late in the election regardless of the letter's release. If this were the case, we might expect to see these same results when treatment is defined around the final presidential debate, October 19th. However, as shown in Section A.5, the results do *not* hold when I redefine treatment in this way. This suggests that the difference-in-differences design identifies the effect of the Comey letter.

3.3 Nevada and North Carolina

A potential issue with the above design is comparing voters across states: several other statewide races coincide with presidential elections, which could lead to confounding influences on turnout and partisan vote shares. To address this concern, I focus solely on two

states with early voting, Nevada and North Carolina, which report early vote turnout by party registration for each week (or in North Carolina, each day). I use this disaggregated data on partisan early voting turnout to compare pre-October 28th early voters to post-October 28th early voters, as only the latter group was able to incorporate information from Comey’s letter into their vote in 2016.

The unique timing of early voting and systems of reporting early voting turnout in Nevada and North Carolina grants me the ability to make within-state comparisons and compare early voters to other early voters. That is, this method rules out confounding cross-state and cross-voting method differences. Additionally, studying turnout by party registration provides insight into the mechanisms behind the results from equation (1); a decrease in turnout by registered Democrats (or an increase in turnout by registered Republicans) would indicate changes along the extensive, rather than intensive, margin. Although they come at the cost of examining only two states, these are valuable contributions.

Nevada had a thirteen-day early voting period that lasted between October 22 and November 4 in 2016; North Carolina had a seventeen-day early voting period, which took place between October 20 and November 5 in 2016. Early voting turnout data is provided by week in Nevada and by day in North Carolina. For consistency, I aggregate the data as follows: early voting turnout prior to October 28th (“Period 1”) and early voting turnout after October 28th (“Period 2”).¹⁷

Using period-county-year observations, I estimate difference-in-differences specifications separately for each state, comparing Period 1 and Period 2 voters in 2016 and in 2012. The identifying assumption here is that, absent the Comey letter, Period 1 and Period 2 voters would have trended similarly in terms of the outcomes of interest. Again, this design also requires that any late-election information in 2012 did not affect political behavior in Nevada

¹⁷In Nevada, Period 1 includes turnout on 10/28 as data is only provided by week. I omit 10/28 turnout in North Carolina from my analysis.

and North Carolina. I estimate

$$y_{pct} = \alpha + \delta T_t + \gamma P_p + \beta(T_t \times P_p) + \epsilon_{pct}, \quad (2)$$

where p indexes early voting period, c indexes county, and t indexes year (it may be helpful to think of the 2016 election as treated and 2012 as control, where Period 2 is the post-period and Period 1 is the pre-period). Outcomes of interest are turnout by party registrants (i.e., voters registered as Democrats, etc.) as a share of total early votes. Unfortunately, election outcomes (i.e., Democratic presidential vote share, etc.) are not provided at this level of disaggregation. However, turnout by party registration is helpful in determining how different “types” of voters reacted along the extensive margin, something I am unable to determine by solely studying election outcomes.

3.4 Early Voting and Election Day Vote Totals

To further address the concern that my main specification does not identify the effect of the letter but instead differential trends in early voting and non-early voting states, I make use of another specification which considers only early voting states. Six early voting states (Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina) report votes cast on Election Day vs. during the early voting period separately, which allows me to compare partisan vote shares of early voters and Election Day voters within those states.

Since the sample is comprised only of early voting states, this method rules out confounding early voting vs. non-early voting state differences. I compare partisan vote shares from early voting and Election Day in 2012 and 2016, considering Election Day as the post-period and the 2016 election as treated. It should be noted that early voting extended past 10/28 in each of these states. However, I still expect to observe the Comey effect so long as sufficiently many early voters cast their ballot ahead of 10/28 in 2016, possessing no knowledge of the

letter when they did so.

For this reason, I expect to observe larger effects in states that had more (vs. less) days of early voting available ahead of the letter’s release. Therefore, I consider a difference-in-differences specification with continuous treatment, leveraging the difference between counties with many vs. few days of early voting prior to 10/28. In addition to the traditional parallel trends assumption, this specification requires a further “strong parallel trends” assumption [Callaway et al., 2024]. In particular, I must assume that counties with many days of early voting prior to 10/28 would have experienced similar effects to counties with few days of early voting prior to 10/28 had they received a “lower dose” (fewer early voting days ahead of the letter’s release) instead.

I estimate

$$y_{pcst} = \alpha + \delta T_t + \gamma P_p + \eta F_s + \beta(D_{st} \times T_t \times P_p) + \epsilon_{pcst}, \quad (3)$$

where T_t takes a value of 1 in the 2016 election and 0 otherwise, and P_p takes a value of 1 for votes cast on Election Day and 0 for votes cast in early voting periods. The “dose,” D_{st} , is the number of days early voting was available prior to 10/28. This is determined independently by each state and only enters the specification in 2016. Finally, I include state fixed effects, F_s , to control for any other variables which vary at the state level. Outcomes of interest are Democratic and Republican presidential vote shares.

4 Results

Table 2 shows results from the main empirical design. First, note that there is no change in turnout. Since there is not a significant change on the extensive margin, I look for changes

on the intensive margin by examining the effect of the Comey letter on presidential vote shares (total ballots cast for a candidate, divided by total presidential ballots cast). The results in Table 2 show an increase in Republican presidential vote share of 3.2 percentage points, in combination with a decrease in Democratic presidential vote share of 2 percentage points.

Table 2 also displays an increase in average Republican down-ballot vote share of 3.2 percentage points and a complementary decrease in average Democratic down-ballot vote share of 3.3 percentage points. As mentioned above, I drop uncontested races from my analysis, explaining the difference in the number of observations across columns. In Section 5, I discuss interpretation of these magnitudes.

Results are robust to the inclusion of time-varying county economic and demographic controls including median income, unemployment rate, population density, median age, number of men per 100 women, percent non-white, and percent of those 25 and up with a Bachelor’s degree or higher (right-hand columns under each outcome). Additionally, results are robust to a refinement considering only counties in Florida and Oklahoma (which had early voting, but not before 10/28) as treated units (Section A.2), dropping counties in states most affected by Hurricane Sandy in 2012 (Section A.3), and excluding counties in states with early voting open before 10/7 (Section A.4). Results do not hold if treatment is redefined around the date of the last presidential debate, 10/19, as opposed to 10/28, suggesting the specification identifies the effect of the Comey letter and not an inevitable late-election downturn for Democrats (Section A.5).

Table 2: Voters most able to respond to the Comey letter were more likely to vote for Republican candidates in presidential and down-ballot races.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	0.009 (0.006)	0.009 (0.007)	-0.02* (0.011)	-0.024 (0.015)	0.032*** (0.01)	0.036** (0.015)	-0.033** (0.015)	-0.035* (0.018)	0.032* (0.017)	0.033* (0.018)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.56		0.347		0.618		0.358		0.608	
Controls	✓		✓		✓		✓			

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-level difference-in-differences specification, weighted by 18+ population: $y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ct}$, where T_c takes a value of 1 in counties *without* early voting open before 10/28/16 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, and percent of those 25 and up with a Bachelor's degree or higher. I calculate turnout rate as a proportion of 18+ population; presidential vote shares are proportions of total presidential votes cast. I calculate U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in the respective races before averaging the three together to calculate the down-ballot vote share. I drop observations with uncontested races, leading to the different number of observations for the down-ballot outcomes.

Table 3 shows results from the complementary Nevada and North Carolina design. In North Carolina, results show a decrease of 2.3 percentage points in the share of early ballots cast by registered Democrats and an increase of 1.9 percentage points in the share of early ballots cast by registered Republicans. In Nevada, results show a decrease of 0.9 percentage points in the share of early ballots cast by registered Democrats. This may serve as informative of who reacted to the Comey letter: it is plausible that the letter mobilized right-leaning and independent voters in Nevada and North Carolina and discouraged left-leaning ones.

Table 3: After the Comey letter was sent, the share of early ballots cast by registered Democrats in Nevada and North Carolina decreased, and the Republican early voting share in North Carolina increased.

	Registered Dem. Early Turnout Share	Registered Rep. Early Turnout Share	Registered Ind. Early Turnout Share
Panel A: North Carolina			
Treat $\times$ Post	-0.023*** (0.003)	0.019*** (0.003)	0.004** (0.002)
N	400	400	400
Mean	0.467	0.312	0.221
Panel B: Nevada			
Treat $\times$ Post	-0.009* (0.006)	0.001 (0.008)	0.009 (0.007)
N	68	68	68
Mean	0.277	0.532	0.191

Standard errors are clustered at the county level and listed in parentheses; standard errors in Panel B are bootstrapped.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-week level difference-in-differences specification, weighted by 18+ population: $y_{pct} = \alpha + \delta T_t + \gamma P_p + \beta(T_t \times P_p) + \epsilon_{pct}$, where T_t takes a value of 1 in 2016 and 0 in 2012, and P_p takes a value of 1 for “Period 2” voters and 0 for “Period 1” voters. Registered Democrat early turnout share is the number of registered Democrats who cast a ballot early divided by the number of early ballots cast; registered Republican and Independent early turnout shares are defined similarly for registered Republicans and voters not registered as Republicans or Democrats, respectively.

Table 4 shows results from the design comparing partisan vote shares from early voting periods versus Election Day. Results indicate that knowledge of the letter increased the Republican presidential vote share and decreased the Democratic presidential vote share. In particular, I estimate that a county with 10.17 days of early voting ahead of the letter’s release (the average number of days of early voting prior to 10/28 across all states with no-excuse early voting in 2016) saw a decrease of 2.03 percentage points in Democratic presidential vote share and an increase of 1.02 percentage points in Republican presidential vote share in Election Day voters relative to early voters. As this specification uses vote shares from early voting states only, it may serve to alleviate the concern that the results

from Table 2 simply reflect differing trends among early voting and non-early voting states.

Table 4: Voters most able to respond to the Comey letter were less likely to vote for the Democratic presidential candidate and more likely to vote for the Republican presidential candidate.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share	
# Days of EV $\times$ Treat $\times$ Post	-0.002*** (0.001)	-0.002*** (0.001)	0.001** (0.001)	0.001* (0.001)
<i>N</i>	2072		2072	
Mean	0.376		0.596	
Controls	✓		✓	

Standard errors are clustered at the county level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-period level difference-in-differences specification, weighted by 18+ population: $y_{pcst} = \alpha + \delta T_t + \gamma P_p + \eta F_s + \beta(D_{st} \times T_t \times P_p) + \epsilon_{pcst}$, where T_t takes a value of 1 in 2016 and 0 in 2012, and P_p takes a value of 1 in the Election Day voting period and 0 in the early voting period. State fixed effects are given by F_s , and D_{st} gives the number of days of early voting available before October 28 in state s in year t . I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, and percent of those 25 and up with a Bachelor’s degree or higher. Presidential vote shares are proportions of total presidential votes cast.

5 Interpreting Magnitudes: An Instrumented Difference-in-Differences Approach

In Table 2, I find that exposure to the Comey letter led to significant decreases in down-ballot Democratic vote shares (and increases in their Republican counterparts). These findings are suggestive of presidential coattail effects: the popularity of a presidential candidate affected the performance of her party’s candidates in concurrent races. Scholars have long sought to estimate presidential coattail effects. However, this is often a difficult task: rarely can the popularity of a presidential candidate be separated from the general attitudes toward her political party.

In order to estimate presidential coattail effects in my setting, I turn to an instrumented difference-in-differences (DDIV) model [Hudson et al., 2017, Duflo, 2001]. This model scales the difference-in-differences effect of the Comey letter on down-ballot partisan vote shares by its effect on partisan presidential vote shares. In other words, it allows me to obtain the effect of a one percentage point increase in partisan presidential vote share, rather than the effect of exposure to the Comey letter itself. The DDIV coefficient β is estimated via the following instrumental variables system:

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta \widehat{Y}_{ct} + \epsilon_{ct} \quad (4)$$

$$Y_{ct} = \theta + \rho T_c + \pi P_t + \eta(T_c \times P_t) + \mu_{ct}, \quad (5)$$

where T_c and P_t are defined as in equation (1). Here, exposure to the Comey letter instruments for Y_{ct} , the Republican presidential vote share in county c in year t .^{18,19} In the second-stage equation, outcomes of interest y_{ct} include average vote shares for Republican and Democratic state-level candidates. I weight by 18+ population in both estimations.

Note that the first-stage parameter, η from equation (5), is equivalent to the coefficient estimated by equation (1) when regressing on presidential vote shares.²⁰ The DDIV coefficient β can be written as the ratio of the reduced-form and first-stage parameters. The reduced-form parameter can be found by estimating:

$$y_{ct} = \nu + \tau T_c + \xi P_t + \phi(T_c \times P_t) + \omega_{ct}, \quad (6)$$

¹⁸I instrument for Republican presidential vote share rather than Democratic for two reasons: a stronger first-stage and ease of interpretation (exposure to the Comey letter increased the Republican – not Democratic – presidential vote share).

¹⁹In the formal treatment of DDIV provided in Hudson et al. [2017], Y_{ct} is assumed to be “a discretely- and positively-valued treatment.” However, the authors note that it is straightforward to extend to a setting with continuous treatment – such as a presidential vote share.

²⁰For the IV exercise, I drop all counties which only had one major party run in a state-level race. This results in a sample of 5400, rather than 5782, observations, and means that η from equation (5) and β from equation (1) differ slightly.

i.e. equation (1) when regressing on state-level candidates' vote shares.

As discussed in Hudson et al. [2017], the DDIV design requires several assumptions. Just as in the difference-in-differences model estimated in equation (1), the assumption of parallel trends must hold (see Section 3.2.1 for discussion). Similar to traditional instrumental variables models, monotonicity and the exclusion restriction must hold. In my context, this requires assuming 1) exposure to the Comey letter did not cause an increase in Democratic presidential vote share for some subpopulations and 2) knowledge of the Comey letter did not affect partisan down-ballot vote shares except through its effect on the popularity of each party's presidential candidates.

Table 5 presents the results from equations (4) and (5) where the outcomes of interest are down-ballot partisan vote shares (averaged over U.S. House, Senate, and gubernatorial elections).²¹ Results indicate that a one percentage point increase in the Republican presidential vote share leads to a 0.86 percentage point increase in the down-ballot Republican vote share.

²¹Note that much of the coattails literature is dedicated to estimating the extent of spillovers between presidential and U.S. House races. In Table A7, I re-estimate equations (4) and (5) using partisan U.S. House vote shares as the sole outcomes of interest.

Table 5: An increase in the Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) down-ballot vote shares.

	(1)	(2)
Rep. Down-Ballot Vote Share	0.863*** (0.332)	0.861*** (0.32)
Dem. Down-Ballot Vote Share	-0.892*** (0.3)	-0.899*** (0.297)
First-Stage (Rep. Pres. Vote Share)	0.037*** (0.014)	0.038** (0.016)
N	5400	5400
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from an instrumental variables system. In the first-stage, I estimate $Y_{ct} = \theta + \rho T_c + \pi P_t + \eta(T_c \times P_t) + \mu_{ct}$, where T_c takes a value of 1 in counties *without* early voting open before 10/28/16 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. Here, Y_{ct} is the Republican presidential vote share (as a proportion of total presidential votes cast) in county c in year t . Then, I estimate $y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta \widehat{Y}_{ct} + \epsilon_{ct}$, where y_{ct} is the average of Republican or Democratic U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in their respective races. All specifications are weighted by 18+ population. In column (2), I include time-varying controls: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, and percent of those 25 and up with a Bachelor's degree or higher. I drop observations where only one major party competed in all relevant races.

6 Conclusion

This paper is the first to make use of an October surprise to estimate presidential coattail effects: using an instrumented difference-in-differences design, I find that a one percentage point increase in Republican presidential vote share leads to a 0.86 percentage point increase in Republican down-ballot vote share. Results in this paper indicate that James Comey's letter to Congress, released just eleven days ahead of the 2016 presidential election, increased both the presidential and down-ballot vote shares of Republican candidates. Using cross-state variation in the availability of early voting, I find that exposure to the letter led to an increase in the Republican presidential vote share of 3.2 percentage points – larger than the

eight closest state margins that election.

Complementary empirical designs suggest that the effect of the letter is robust to limiting to both within-state and early voting state-only comparisons. In a specification comparing partisan vote shares among early voters and Election Day voters in early voting states, I find a differential increase in Republican presidential vote share in counties with more early voting days (vs. less) ahead of the letter’s release. Evidence suggests that at least part of these effects are owed to changes in partisan turnout: when comparing pre-10/28 early voters to post-10/28 early voters, I find that exposure to the letter led to a decrease in registered Democrats’ early turnout share of 2.3 percentage points in North Carolina and 0.9 percentage points in Nevada.

The findings of this paper indicate that the popularity or electoral strength of a party’s presidential candidate has meaningful downstream effects. This raises numerous strategic considerations for down-ballot political candidates: for example, how closely to align with prominent co-partisan candidates or even when to run for office. These spillovers have important implications for a party’s agenda, too: presidential coattails imply that the more popular a newly-elected president is, the more co-partisans she can expect to be seated in her new Congress. The finding that late-election information regarding a presidential candidate has down-ballot effects suggests that there are strong incentives for the strategic release of information during presidential campaigns beyond first-order effects on the presidential race.

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A Appendix

This appendix contains robustness checks for the identification designs presented in the paper and additional results from the DDIV model. Sections [A.1](#), [A.2](#), [A.3](#), [A.4](#), and [A.5](#) present robustness checks for the main research design, introduced in Section [3.2](#). Section [A.6](#) gives results from the DDIV specification using only partisan U.S. House vote shares as the outcome.

A.1 Balance Test

In equation [\(1\)](#), I leverage cross-state variation in preexisting convenience voting laws at the time of the Comey letter. Counties in states with early voting open prior to 10/28 are controls, and counties in states 1) with early voting open only *after* 10/28 and 2) with no early voting at all are treated. The identifying assumption is the parallel trends assumption: absent the Comey letter, treatment and control counties should have trended similarly in the outcomes of interest. However, one might be concerned that states with and without early voting might have been on different political trends between 2012 and 2016, violating this assumption.

I conduct a balance test in order to address this concern. In particular, I examine whether treatment is related to items on states' ballots whose presence may reflect varying political trends or a differential reason to turn out or to vote for a specific partisan candidate in that election year. I focus on four broad categories: number (and type) of ballot initiatives, toss-up Senate or gubernatorial races, the availability of straight-ticket voting, and the presence of a third-party presidential candidate.

Ballot initiatives have been shown to increase voter turnout [[Tolbert et al., 2001](#)]; for this reason, I test whether the number of ballot initiatives is related to treatment. Different cat-

egories of ballot measures can serve to mobilize specific groups of voters [Garretson, 2014], potentially leading to spillover effects on other election outcomes. For example, Gardner et al. [2025] find that the presence of abortion-related ballot measures boosted Democrats’ performance in congressional races in the 2022 midterm elections. I collect data from Ballotpedia on a few major categories of ballot measures (taxes, healthcare, labor, social issues) to test for a relationship between the presence of different ballot initiative categories and treatment.^{22,23}

Especially competitive races at the state level might increase voter turnout, amplify partisan campaigning efforts, etc. I collect data on which states had toss-up Senate and gubernatorial races in 2012 and 2016, as defined by the Cook Political Report around three weeks ahead of each election. Third-party presidential candidates were particularly important in the 2016 election, so I also tested whether the presence of third-party presidential candidates on the ballot can be predicted by treatment.

Since I leverage exposure to the Comey letter to estimate coattail effects, I am especially concerned with voters’ access to straight-ticket voting, which should theoretically increase the strength of such effects (as it makes voting for a single party up and down the ballot particularly easy). In fact, several states did away with straight-ticket voting in the years preceding 2016. I use information from the National Conference of State Legislatures to code each state’s straight-ticket status in 2012 and 2016.

I re-estimate equation (1) and place indicator variables on the left-hand side for each of these variables (or for number of ballot measures, number of ballot measures divided by ten). Results are shown in Figure A1. There is no statistically significant relationship between treatment and any of the covariates, supporting the parallel trends assumption.

²²In the 2012 presidential election, four states had ballot initiatives aimed at restricting the Affordable Care Act. I put these ballot measures in their own category separate from other healthcare-related initiatives.

²³The “social issues” category includes measures on gun rights, same-sex marriage, abortion, the death penalty, and marijuana legalization.

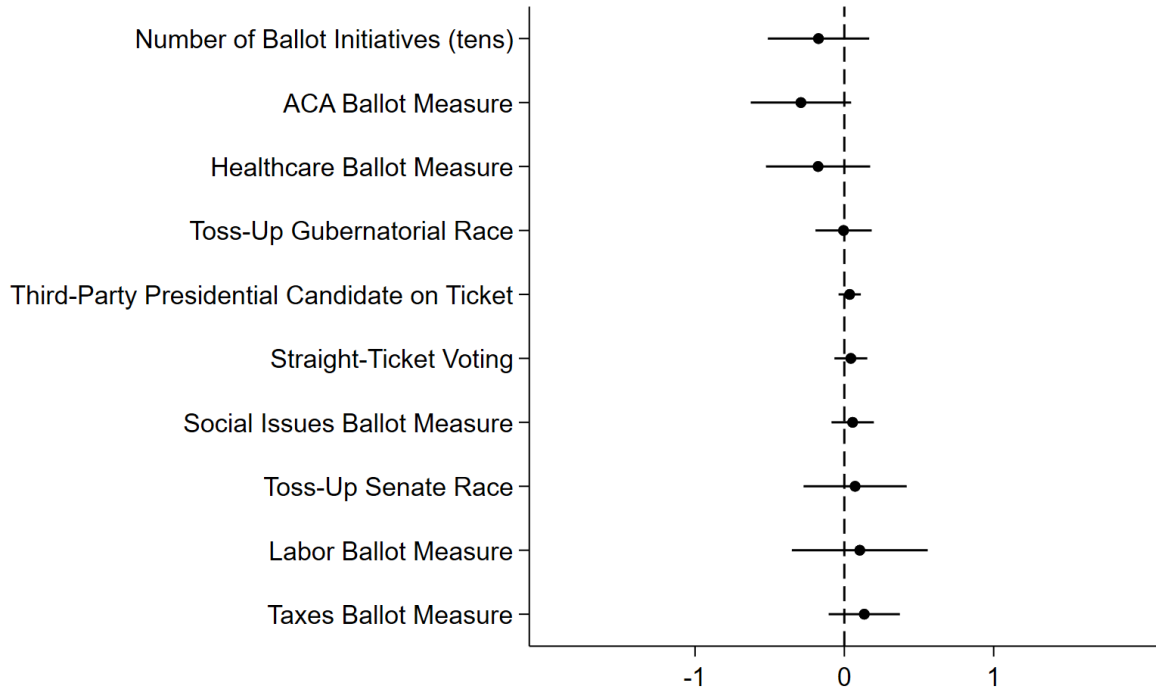


Figure A1: There is no effect of “treatment” on states’ ballots. That is, early voting and non-early voting states did not appear to be on different political trends between 2012-2016. Notes: this figure displays results from variations of my main specification which place ballot characteristics on the left-hand side.

A.2 Florida and Oklahoma

In this section, I make a refinement to the design to further address the “differential trends” concern: I limit treatment counties to those in group (1): with early voting open only *after* 10/28 (all counties in Florida and Oklahoma). Then, I re-estimate (1) with the refined treatment group and same control group (all counties in states with early voting open prior to 10/28). This allows me to make direct comparisons between early voting states *only*, rather than relying on comparisons between early voting and non-early voting states.

Results are in Table A1. Here, there *is* a change on the extensive margin: voters in Florida and Oklahoma, who were more able to respond to the Comey letter, saw an increase

in turnout rate of 1.5 percentage points. This is a bit puzzling: if the control group already has a number of voters “locked in” and the rest (who did not vote early) are free to respond, then an increase in turnout does not make much sense. This finding persists when limiting the control group to counties in neighboring states (Arkansas, Georgia, Kansas, New Mexico, and Texas; see Table A2). However, looking at disaggregated turnout of a subset of these neighbors in Table A3, I show that the Comey letter did not lead to a significant change in early turnout rate or Election Day turnout rate. This result appears noisy and is perhaps due to measurement error in the denominator.²⁴

Returning to Table A1, the results show an increase in Republican presidential and state-level vote shares. These results hold (and strengthen) when comparing Florida and Oklahoma only to their neighboring states (see Table A2). Broadly, the results remain consistent with those found in my main specification and reported in Table 2.

Table A1: Voters in Florida and Oklahoma were more likely to vote for Republican presidential and down-ballot candidates than voters in states with convenience voting open before 10/28/16.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.015*** (0.004)	0.012* (0.006)	-0.006 (0.01)	-0.007 (0.017)	0.017* (0.01)	0.017 (0.015)	-0.045*** (0.013)	-0.035* (0.018)	0.052*** (0.015)	0.033* (0.018)
N	4180		4180		4180		3814		3814	
Mean	0.555		0.334		0.629		0.341		0.621	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 using only counties in Florida and Oklahoma as treatment units.

²⁴If 18+ population were growing more quickly in Florida than documented by the SEER measures, this could lead to a false finding of increased turnout rate.

Table A2: Voters in Florida and Oklahoma were more likely to vote for Republican presidential and down-ballot candidates than voters in neighboring states with convenience voting open before 10/28/16.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.012** (0.005)	0.01 (0.009)	-0.03* (0.017)	-0.01 (0.039)	0.031*** (0.012)	0.011 (0.03)	-0.05 (0.04)	-0.008 (0.05)	0.076*** (0.041)	0.045 (0.034)
<i>N</i>	1288		1288		1288		1288		1288	
Mean	0.507		0.298		0.674		0.292		0.665	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 using only counties in Florida and Oklahoma as treatment units and counties in their neighboring states as controls.

Table A3: The Comey letter led to no significant change in the Election Day or early turnout rates in Florida and Oklahoma counties, as compared to counties in their neighboring states: Arkansas, Georgia, and New Mexico.

	Turnout Rate		Early Turnout Rate		Election Day Turnout Rate	
Treat×Post	0.017*** (0.006)	0.018** (0.008)	0.015 (0.026)	0.017 (0.034)	-0.008 (0.027)	-0.01 (0.039)
<i>N</i>	816		816		816	
Mean	0.512		0.2		0.266	
Controls	✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table disaggregates results from Table A2, examining the impact of the letter on turnout by method. Note that the control sample differs from that in Table A2: Kansas and Texas do not provide turnout by voting method and are thus dropped from the estimation.

A.3 Hurricane Sandy

The first empirical design, summarized in equation (1), takes 2012 as a base year for the difference-in-differences specification. Thus, I assume that any late-election information in 2012 did not affect election outcomes. One potential violation of this assumption comes

from Hurricane Sandy, a tropical cyclone which made landfall in the United States around October 29, 2012. Many at the time posited the storm would benefit incumbent Democrat President Obama, who was widely praised for his response. However, as found in Hart [2014], the effect of Hurricane Sandy on the election was “variable and small in magnitude.”

To account for any possible impact of Hurricane Sandy, I re-estimate equation (1) after omitting counties in Connecticut, New Jersey, and New York. These three states had the largest populations in storm surge zones during Hurricane Sandy (Center for International Earth Science Information Network [2012]).²⁵ As such, it is probable that any effect Hurricane Sandy had on political outcomes was largest there. Results are robust to this omission; see Table A4.

Table A4: Results are robust to the omission of Connecticut, New Jersey, and New York.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.005 (0.007)	0.005 (0.007)	-0.02* (0.012)	-0.025 (0.017)	0.032*** (0.011)	0.036** (0.017)	-0.043*** (0.014)	-0.046*** (0.017)	0.043** (0.016)	0.045** (0.017)
<i>N</i>	5600		5600		5600		5218		5218	
Mean	0.561		0.342		0.623		0.351		0.614	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after omitting counties in Connecticut, New Jersey, and New York from the analysis.

A.4 October 7th, 2016

As described in Section 2.2, October 7th was an important day in the 2016 presidential election. Three major events unfolded: *The Washington Post* reported on the 2005 “Access Hollywood” tape, the U.S. government accused Russia of meddling in the election, and WikiLeaks began releasing emails of John Podesta. By 10/7, ten states had already begun early voting: Idaho, Iowa, Maine, Nebraska, New Jersey, Ohio, South Dakota, Vermont,

²⁵Along with Massachusetts, which is already excluded from my analysis.

Virginia, and Wyoming. In order to ensure the results from (1) are not driven by comparing pre-10/7 voters to post-10/7 voters, I omit these ten states from my analysis below. Results are robust to this omission; see Table A5.

Table A5: Results are robust to the omission of counties which had access to early voting prior to October 7th, 2016.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.009 (0.006)	0.009 (0.007)	-0.031*** (0.01)	-0.035** (0.016)	0.043*** (0.009)	0.047*** (0.016)	-0.043*** (0.014)	-0.045** (0.018)	0.046*** (0.016)	0.048*** (0.017)
<i>N</i>	4588		4588		4588		4281		4281	
Mean	0.546		0.344		0.622		0.356		0.607	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after excluding states with early voting open before 10/7/16 from the analysis.

A.5 October 19th, 2016

It is possible that the central empirical design simply captures changing dynamics throughout the campaign leading up to the election. It could be the case that Clinton and the Democratic party would have performed worse with “later” versus “earlier” voters, even in the absence of the letter. To address this concern, I redefine treatment around October 19th, 2016, the date of the last presidential debate. If I find that voters reacted to the October 19th “treatment”, then it is likely that the results in Table 2 are reflective of voter preferences shifting over time in response to a number of factors rather than the singular Comey letter “shock.” As can be seen in Table A6, I do *not* find that voters responded to this false treatment, increasing confidence in the attribution of my results to the Comey letter.

Table A6: The results shown in Table 2 do not hold when treatment is redefined around October 19th, 2016, suggesting the results were due to the Comey letter and not simply the larger dynamics of the election.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.001 (0.006)	0.005 (0.007)	-0.006 (0.016)	-0.015 (0.021)	0.011 (0.016)	0.02 (0.021)	-0.005 (0.02)	-0.016 (0.023)	0.002 (0.021)	0.011 (0.022)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.56		0.347		0.618		0.358		0.608	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after re-defining treatment: counties *without* early voting open before 10/19/16 – the date of the last presidential debate – are considered treated; those with early voting open ahead of 10/19/16 are controls.

A.6 Instrumented Difference-in-Differences Results on U.S. House Vote Shares

The results in Table 5 demonstrate the impact of a one percentage point increase in Republican presidential vote share on partisan down-ballot vote shares. I calculate these down-ballot vote shares by calculating the average of Republican or Democratic U.S. House, Senate, and gubernatorial vote shares. However, much of the presidential coattails literature uses only U.S. House vote shares as outcomes – likely because, unlike senators and governors, U.S. House representatives face re-election every two years.

Table A7 presents results from the DDIV model where the outcomes of interest are partisan U.S. House vote shares. I find that a one percentage point increase in Republican presidential vote share leads to a 0.51 percentage point increase in Republican U.S. House vote share.

Table A7: An increase in Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) U.S. House vote shares.

	(1)	(2)
Rep. U.S. House Vote Share	0.514** (0.254)	0.549** (0.223)
Dem. U.S. House Vote Share	-0.49* (0.291)	-0.527* (0.303)
First-Stage (Rep. Pres. Vote Share)	0.035*** (0.013)	0.037** (0.016)
N	4910	4910
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 5 using only U.S. House vote shares as the outcome of interest. I drop observations where only one major party competed.